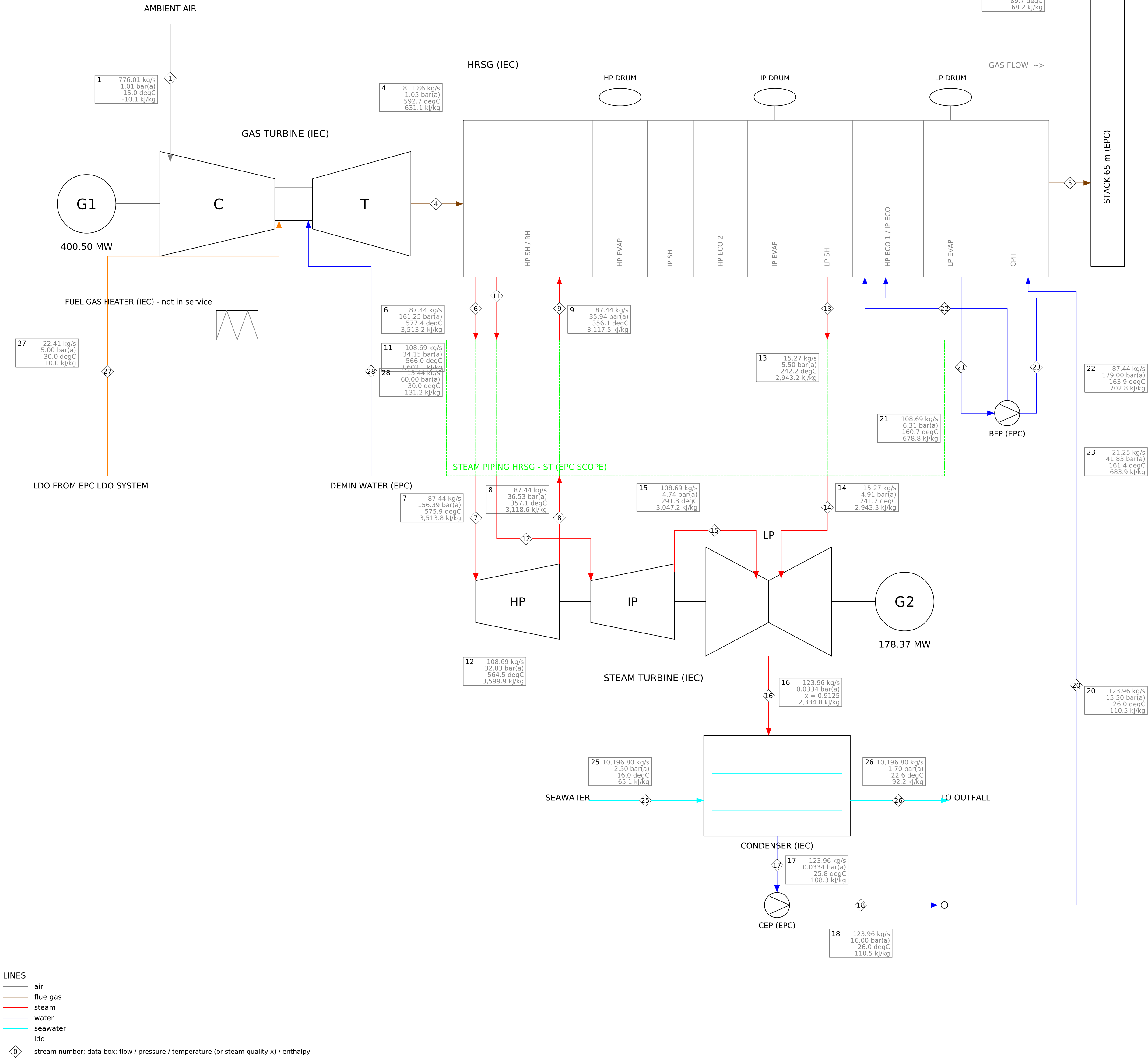




STREAM TABLE - SRC-LDO-100

| No | Stream                                                   | kg/s      | bar(a) | degC / x | kJ/kg   | MW*   |
|----|----------------------------------------------------------|-----------|--------|----------|---------|-------|
| 1  | Ambient air to GT compressor inlet                       | 776.01    | 1.01   | 15.0     | -10.1   | -7.8  |
| 4  | GT exhaust gas to HRSG inlet                             | 811.86    | 1.05   | 592.7    | 631.1   | 512.4 |
| 5  | Flue gas at HRSG outlet to stack                         | 811.86    | 1.01   | 89.7     | 68.2    | 55.4  |
| 6  | HP steam at HRSG outlet (HP SH outlet)                   | 87.44     | 161.25 | 577.4    | 3,513.2 | 307.2 |
| 7  | HP steam at ST main stop valve inlet                     | 87.44     | 156.39 | 575.9    | 3,513.8 | 307.2 |
| 8  | Cold reheat steam at HP turbine exhaust                  | 87.44     | 36.53  | 357.1    | 3,118.6 | 272.7 |
| 9  | Cold reheat steam at HRSG reheater inlet                 | 87.44     | 35.94  | 356.1    | 3,117.5 | 272.6 |
| 10 | IP superheated steam (joins cold reheat inside the HRSG) | 21.25     | 37.26  | 312.6    | 3,004.7 | 63.9  |
| 11 | Hot reheat steam at HRSG outlet                          | 108.69    | 34.15  | 566.0    | 3,602.1 | 391.5 |
| 12 | Hot reheat steam at ST IP stop/control valve inlet       | 108.69    | 32.83  | 564.5    | 3,599.9 | 391.3 |
| 13 | LP steam at HRSG outlet                                  | 15.27     | 5.50   | 242.2    | 2,943.2 | 44.9  |
| 14 | LP steam at ST LP admission                              | 15.27     | 4.91   | 241.2    | 2,943.3 | 44.9  |
| 15 | IP turbine exhaust (crossover, internal)                 | 108.69    | 4.74   | 291.3    | 3,047.2 | 331.2 |
| 16 | LP turbine exhaust to condenser (UEEP)                   | 123.96    | 0.0334 | x 0.913  | 2,334.8 | 289.4 |
| 17 | Condensate at hotwell outlet / CEP suction               | 123.96    | 0.0334 | 25.8     | 108.3   | 13.4  |
| 18 | Condensate at CEP discharge                              | 123.96    | 16.00  | 26.0     | 110.5   | 13.7  |
| 20 | Condensate to HRSG condensate preheater inlet            | 123.96    | 15.50  | 26.0     | 110.5   | 13.7  |
| 21 | Feedwater from LP drum to boiler feed pumps              | 108.69    | 6.31   | 160.7    | 678.8   | 73.8  |
| 22 | HP feedwater at BFP discharge to HP economiser           | 87.44     | 179.00 | 163.9    | 702.8   | 61.5  |
| 23 | P feedwater at BFP interstage to IP economiser           | 21.25     | 41.83  | 161.4    | 683.9   | 14.5  |
| 25 | Circulating seawater to condenser inlet                  | 10,196.80 | 2.50   | 16.0     | 65.1    | 663.9 |
| 26 | Circulating seawater from condenser outlet               | 10,196.80 | 1.70   | 22.6     | 92.2    | 939.9 |
| 27 | LDO at liquid fuel skid inlet (IEC terminal point IF-02) | 22.41     | 5.00   | 30.0     | 10.0    | 0.2   |
| 28 | Demineralised water for NOx water injection (IF-03)      | 13.44     | 60.00  | 30.0     | 131.2   | 1.8   |

\* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

|                                       |        |        |
|---------------------------------------|--------|--------|
| GT output (generator terminals)       | 400.50 | MW     |
| ST output (generator terminals)       | 178.37 | MW     |
| Gross output                          | 578.87 | MW     |
| Auxiliary loads                       | -9.35  | MW     |
| Transformer losses (GSU, UAT, MV/LV)  | -1.73  | MW     |
| NET OUTPUT (HV terminals)             | 567.80 | MW     |
| Heat input (LHV)                      | 961.22 | MW     |
| Gross heat rate (LHV)                 | 5,978  | kJ/kWh |
| NET HEAT RATE (LHV)                   | 6,094  | kJ/kWh |
| Net efficiency (LHV)                  | 59.07  | %      |
| Fuel LHV (ISO 6976, from composition) | 42.90  | MJ/kg  |
| Circulating water flow                | 36,112 | m3/h   |
| GT load                               | 100.00 | %      |
| PLANT LOAD (NET, % OF SRC-NG-100)     | 92.48  | %      |
| Water injection (NOx)                 | 13.44  | kg/s   |

CHECKS (guarantees, limits, balances)

|    |                                                                                                                                                       |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| OK | energy balance CMX: residual +0.001 MW (tolerance 0.25 MW)                                                                                            |
| OK | energy balance COND: residual +0.047 MW (tolerance 0.95 MW)                                                                                           |
| OK | energy balance GT: residual +0.016 MW (tolerance 0.92 MW)                                                                                             |
| OK | energy balance HRSG: residual +1.860 MW vs stated casing loss 1.828 MW (tolerance 0.87 MW)                                                            |
| OK | energy balance PIPE: residual +0.286 MW (tolerance 1.02 MW)                                                                                           |
| OK | energy balance ST: residual -0.002 MW (tolerance 0.74 MW)                                                                                             |
| OK | CW temperature rise 6.65 K vs DP-CASE-0015 <= 7.0 K (Condenser cooling water temperature rise)                                                        |
| OK | net export 567.8 MW vs DP-OWNR-0051 connection capacity 640 MW (this case)                                                                            |
| OK | gross output 578.87 MW vs IG-09 >= 572.00 (margin +1.20 %)                                                                                            |
| OK | gross heat rate 5,977.82 kJ/kWh vs IG-10 <= 6,020.00 (margin +0.70 %)                                                                                 |
| OK | net output 567.80 MW vs PG-15 >= 560.00 (margin +1.39 %)                                                                                              |
| OK | net heat rate 6,094.42 kJ/kWh vs PG-16 <= 6,150.00 (margin +0.90 %)                                                                                   |
| OK | guarantee cover: at IG-09/IG-10 limits and this HMB's auxiliaries the net plant is 560.9 MW / 6,139 kJ/kWh vs Owner PG-15 560 MW / PG-16 6,150 kJ/kWh |

|          |                                                                                  |
|----------|----------------------------------------------------------------------------------|
| PROJECT  | Alpilla 600 MW Combined Cycle Power Plant                                        |
| TITLE    | HEAT AND MASS BALANCE - SRC-LDO-100                                              |
| CASE     | Site Reference Conditions, LDO with water injection, base load (guarantee PG-15/ |
| DOCUMENT | ALP-HMB-SRC-LDO-100                                                              |
| STATUS   | preliminary - generated by engine hmb v1.2.0                                     |
| DATA     | 6b60fce553593bf5   2026-09-29                                                    |
| NOTE     | all values calculated from the database; do not edit                             |